Finding Pareto Stationary Point of MILP by BCD/ADMM

1. Optimization problem we want to solve

$$\begin{aligned}\min_{\boldsymbol{x},\boldsymbol{z}} \quad & \mathbf{1}^\top G\boldsymbol{z} \\ \text{s.t.} \quad & A\boldsymbol{x} \le b, \quad \boldsymbol{x} \in \{0,1\}^N \\ & \boldsymbol{z} \ge E\boldsymbol{x} + \boldsymbol{f} - \mathbf{1} \\ & \boldsymbol{z} \ge \mathbf{0}\end{aligned}$$

1.1 Major characteristics of given MILP problem

1. $$\boldsymbol{x}_s = \begin{bmatrix}\boldsymbol{x}_{s_1} \\ \boldsymbol{x}_{s_2} \\ \vdots \\ \boldsymbol{x}_{s_p}\end{bmatrix} \to \boldsymbol{x} = P\boldsymbol{x}_s$$

   where P is a permutation matrix (given), or index mapping (computable).

2. Decision variables for each agent:

   $$\boldsymbol{x}_{s_i} = \{\boldsymbol{x}_{s_1}, \boldsymbol{x}_{s_2}, \dots, \boldsymbol{x}_{s_p}\}$$

   Feasibility of each agent's decision variables are decoupled, i.e.,

   $$A_{s_1}\boldsymbol{x}_{s_1} \le b_{s_1},\ A_{s_2}\boldsymbol{x}_{s_2} \le b_{s_2},\ \dots, A_{s_p}\boldsymbol{x}_{s_p} \le b_{s_p}.$$

   Therefore, the value of $\boldsymbol{x}_{s_i}$ cannot change the feasible region of other $\boldsymbol{x}_{s_j}$.

3. The feasible region for each element of $\boldsymbol{z}$ will be classified into two cases:

   $$z_i \ge 0 \quad \text{or} \quad z_i \ge 1.$$

4. Each agent's decision can influence the feasible region of $\boldsymbol{z}$.

   Example 1 (one agent changes decison while other's decision remains constant):

   $$\boldsymbol{x}'_{s_1}\text{changes and } \boldsymbol{x}_{s_2}, \cdots, \boldsymbol{x}_{s_p} \text{ remains constant},$$

   $$\boldsymbol{x}' = P\begin{bmatrix}\boldsymbol{x}'_{s_1} \\ \boldsymbol{x}_{s_2} \\ \vdots \\ \boldsymbol{x}_{s_p}\end{bmatrix} \Rightarrow \boldsymbol{z}' \le E\boldsymbol{x}' + \boldsymbol{f} - \mathbf{1} = EP\begin{bmatrix}\boldsymbol{x}'_{s_1} \\ \boldsymbol{x}_{s_2} \\ \vdots \\ \boldsymbol{x}_{s_p}\end{bmatrix} + \boldsymbol{f} - \mathbf{1}$$

   Example 2 (every agent changes their decision simultaneously):

   $$\text{Every agent } \boldsymbol{x}'_{s_1}, \boldsymbol{x}'_{s_2} + \cdots, \boldsymbol{x}'_{s_p} \text{ changes simultaneously},$$

   $$\boldsymbol{x}' = P\begin{bmatrix}\boldsymbol{x}'_{s_1} \\ \boldsymbol{x}'_{s_2} \\ \vdots \\ \boldsymbol{x}'_{s_p}\end{bmatrix} \Rightarrow \boldsymbol{z}' \le E\boldsymbol{x}' + \boldsymbol{f} - \mathbf{1}$$

Changing $\boldsymbol{x}_{s_i} \to \boldsymbol{x}'_{s_i}$ does not influence the feasibility (or doable) of changing $\boldsymbol{x}_{s_j} \to \boldsymbol{x}'_{s_j}$, but multiple simultaneous changes may alter the feasible region of the same element of $\boldsymbol{z}$.

Therefore, we can consider

- sequential decision change
- parallel decision change

5. G is a diagonal matrix with all positive diagonal values.

   - If the feasible region of z_i satisfies $z_i \geq 0 \ \text{ or } \ z_i \geq 1$, then corresponding z_i will be immediately decided to $z_i = 0 \ \text{ and } \ z_i = 1$.
   - Therefore, G functions as a weighting matrix, and no additional computation cost arises if the feasible space of $\boldsymbol{z}$ is defined.

2. Multi-Task Learning Strategy to Solve Given MILP

- BCD (Block Coordinate Descent): Sequential / Stackelberg game approach.
- ADMM (Alternating Direction Method of Multipliers): Parallel / dynamic game approach for task consensus.

2.1 BCD Approach

Each agent s_i decides optimal $\boldsymbol{x}_{s_i}$ while assuming other agents' decisions $\boldsymbol{x}_{s_{-i}}$ remain constant. The cost function should be decreased if the agent changes its decision.

2.1.1 BCD Process

1. Input:

$$\boldsymbol{x}^k = P \begin{bmatrix} \boldsymbol{x}^k_{s_1} \\ \vdots \\ \boldsymbol{x}^k_{s_p} \end{bmatrix}, \quad \boldsymbol{z}^k$$

2. Iteration: solve the following MILP for each agent sequentially

$$\boldsymbol{x}^{k+1}_{s_i} = \arg\min_{\boldsymbol{x}_{s_i}} \mathbf{1}^\top G \boldsymbol{z}$$

$$\text{s.t. } A_{s_i}\boldsymbol{x}_{s_i} \leq b_{s_i}, \quad \boldsymbol{z} \geq EP \begin{bmatrix} \boldsymbol{x}^k_{s_1} \\ \vdots \\ \boldsymbol{x}_{s_i} \\ \vdots \\ \boldsymbol{x}^k_{s_p} \end{bmatrix} + \boldsymbol{f} - \mathbf{1}$$

3. Continue until $\boldsymbol{x}^{k+1}_{s_i} = \boldsymbol{x}^k_{s_i}$ for all s_i.

2.2 ADMM with Heuristic/Surrogate Approach

Each agent computes its optimal decision $\boldsymbol{x}_{s_i}$ and corresponding $\boldsymbol{z}_{s_i}$, assuming other agents' decisions remain constant. The goal is to enforce consensus:

$$\boldsymbol{z}_{s_1} = \boldsymbol{z}_{s_2} = \cdots = \boldsymbol{z}_{s_p} = \boldsymbol{z}$$

2.2.1 ADMM Concensus Optimization Problem

$$\min_{\boldsymbol{z}_{s_i}, \boldsymbol{z}} \sum_{i=1}^{p} \mathbf{1}^\top G \boldsymbol{z}_{s_i}$$
$$\text{s.t. } \boldsymbol{z}_{s_i} = \boldsymbol{z}, \quad \forall i = 1, \dots, p.$$

Augmented Lagrangian for each agent:

$$\mathcal{L}_\rho = \sum_{i=1}^{p} \left[\mathbf{1}^\top G \boldsymbol{z}_{s_i} + \boldsymbol{\lambda}_{s_i}^\top (\boldsymbol{z}_{s_i} - \boldsymbol{z}) + \frac{\rho}{2} \|\boldsymbol{z}_{s_i} - \boldsymbol{z}\|_2^2 \right]$$

2.2.2 ADMM Consensus Process

1. $\boldsymbol{x}_{s_i}, \boldsymbol{z}_{s_i}$ update: for each s_i,

$$(\boldsymbol{x}_{s_i}^{k+1}, \boldsymbol{z}_{s_i}^{k+1}) = \arg\min_{\boldsymbol{x}_{s_i}, \boldsymbol{z}_{s_i}} \mathbf{1}^\top G \boldsymbol{z}_{s_i} + \boldsymbol{\lambda}_{s_i}^\top (\boldsymbol{z}_{s_i} - \boldsymbol{z}^k) + \frac{\rho}{2} \|\boldsymbol{z}_{s_i} - \boldsymbol{z}^k\|_2^2$$

$$\text{s.t. } A_{s_i} \boldsymbol{x}_{s_i} \le b_{s_i}, \quad \boldsymbol{z}_{s_i} \ge EP \begin{bmatrix} \boldsymbol{x}_{s_1}^k \\ \vdots \\ \boldsymbol{x}_{s_i} \\ \vdots \\ \boldsymbol{x}_{s_p}^k \end{bmatrix} + \boldsymbol{f} - \mathbf{1}, \quad \boldsymbol{z}_{s_i} \ge \mathbf{0}$$

2. $\boldsymbol{x}$ update:

$$\boldsymbol{x}^{k+1} = P \begin{bmatrix} \boldsymbol{x}_{s_1}^{k+1} \\ \vdots \\ \boldsymbol{x}_{s_p}^{k+1} \end{bmatrix}$$

3. $\boldsymbol{z}$ update (two options):

   Option (i):

$$\boldsymbol{z}^{k+1} = \max \boldsymbol{z}_{s_i}^{k+1} \quad \text{(element-wise)}$$

   Option (ii):

$$\boldsymbol{z}^{k+1} = \arg\min_{\boldsymbol{z}} \sum_{i=1}^{p} \left[\boldsymbol{\lambda}_{s_i}^\top (\boldsymbol{z}_{s_i}^{k+1} - \boldsymbol{z}) + \frac{\rho}{2} \|\boldsymbol{z}_{s_i}^{k+1} - \boldsymbol{z}\|_2^2 \right]$$
$$\text{s.t. } \boldsymbol{z} \ge E\boldsymbol{x}^{k+1} + \boldsymbol{f} - \mathbf{1}, \quad \boldsymbol{z} \ge 0$$

4. Dual update: for each s_i,

$$\boldsymbol{\lambda}_{s_i}^{k+1} = \boldsymbol{\lambda}_{s_i}^k + \rho(\boldsymbol{z}_{s_i}^{k+1} - \boldsymbol{z}^{k+1})$$

Continue until $\boldsymbol{z}_{s_i} = \boldsymbol{z}$ for all s_i. If iteration stops, it reached to Pareto stationary.

2.2.3 Global z-update (in ADMM-Option (ii))

$$
\begin{aligned}
z^{k+1} &= \arg\min_{\boldsymbol{z}} \sum_{i=1}^{p} \Big[(\lambda_{s_i}^{k})^{\top} (\boldsymbol{z}_{s_i}^{k+1} - \boldsymbol{z}) + \frac{\rho}{2} \|\boldsymbol{z}_{s_i}^{k+1} - \boldsymbol{z}\|_2^2 \Big] \\
\text{s.t.} \quad & \boldsymbol{z} \geq EP\,\boldsymbol{x}^{k+1} + f - \mathbf{1}, \qquad \boldsymbol{z} \geq \mathbf{0}
\end{aligned}
$$

1. Quadratic form (dropping constants).

$$
z^{k+1} = \arg\min_{z} \ \frac{\rho p}{2}\, \boldsymbol{z}^{\top} \boldsymbol{z} - \Big(\sum_{i=1}^{p} \lambda_{s_i}^{k} + \rho \sum_{i=1}^{p} \boldsymbol{z}_{s_i}^{k+1} \Big)^{\top} \boldsymbol{z}
$$

2. Unconstrained minimizer

$$
\boldsymbol{z}_{\text{unproj}}^{k+1} = \frac{1}{p} \sum_{i=1}^{p} \boldsymbol{z}_{s_i}^{k+1} + \frac{1}{\rho p} \sum_{i=1}^{p} \lambda_{s_i}^{k}
$$

   Define averages,

$$
\begin{aligned}
\bar{\boldsymbol{z}}^{k+1} &:= \frac{1}{p} \sum_{i=1}^{p} \boldsymbol{z}_{s_i}^{k+1} \\
\bar{\lambda}^{k} &:= \frac{1}{p} \sum_{i=1}^{p} \lambda_{s_i}^{k}
\end{aligned}
$$

   Then,

$$
\boldsymbol{z}_{\text{unproj}}^{k+1} = \bar{\boldsymbol{z}}^{k+1} + \frac{1}{\rho} \bar{\lambda}^{k}
$$

3. Projection onto the feasibility set

$$
\begin{aligned}
\ell^{k+1} &= \max\big(EP\,\boldsymbol{x}^{k+1} + \boldsymbol{f} - \mathbf{1},\ \mathbf{0} \big) \Rightarrow \text{lowerbound} \\
\boldsymbol{z}^{k+1} &= \Pi_{\boldsymbol{z} \geq \ell^{k+1}} \big(\boldsymbol{z}_{\text{unproj}}^{k+1} \big)
\end{aligned}
$$

4. Final closed-form update (element-wise)

$$
\boxed{\boldsymbol{z}^{k+1} = \max\Big(\bar{\boldsymbol{z}}^{k+1} + \frac{1}{\rho} \bar{\lambda}^{k},\ EP\,\boldsymbol{x}^{k+1} + \boldsymbol{f} - \mathbf{1},\ \mathbf{0} \Big)}
$$